Fast Food Density and Emergency Department Use for Diet-Related Ambulatory Care Sensitive Conditions Among Medicaid-Eligible Populations

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Introduction

Diet-related chronic disease within the United States imposes enormous economic costs, with obesity alone accounting for $173 billion in medical expenditures. Despite existing research linking fast-food access to poorer health conditions, evidence on whether place-based exposure to such outlets results in measurable healthcare remains limited. In a quasi-experimental study of New York City public-housing residents, Han et al. (2020) discovered that moving 0.1 miles closer to a fast-food restaurant increases a child's probability of obesity by 0.6 percentage points. This finding suggests the local food environment can have causal, negative health outcomes for low-income populations, and trigger downstream welfare losses. In this paper, we aim to uncover the relationship between fast-food concentration and emergency department use for diet-related complications among such populations.

Low income Americans are disproportionately exposed to unhealthy food environments, defined as the geographic and economic conditions that shape what foods are accessible and convenient for people in a given community, influencing diet related health outcomes (European Public Health Alliance [EPHA], 2019). Given this definition, there are two commonly cited kinds of food environments: food swamps and food deserts. Food deserts, common in economically disadvantaged neighborhoods in the United States, are geographic areas that have limited access to healthy food stores (National Collaborating Centre for Environmental Health, 2017). Newer work highlights food swamps, where unhealthy food options outnumber healthy food retail and may better predict obesity patterns (Cooksey-Stowers et al., 2017). This debate matters due to the inequity in the distribution of chronic health conditions across different income groups, and national evidence shows lower income populations facing higher risk rates of obesity and long term health and financial burdens from obesity related illness (Wang et al., 2008). Cotti, Gordanier, and Ozturk (2020) found that emergency rooms visits increase near the end of SNAP benefit months which links food insecurity and low/constrained food budgets to an increased number of emergency room visits. These findings lead us to believe that the food environment is not just a public health concern but a policy issue.

Existing research strongly supports a link between fast food access and adverse health outcomes, but it stops short of examining downstream healthcare utilization. For example, in Currie et al. (2010) the use of large scale U.S. data on millions of ninth graders and pregnant

women show that very close proximity to fast food restaurants causally increases obesity rates and excessive pregnancy weight gain. Their quasi-experimental design of comparing individuals within 0.1 miles of a fast food outlet to those slightly farther away addresses selection bias and strengthens the argument that fast food access can directly worsen measurable health indicators. However, their outcome variables are limited to weight related measures. They do not examine whether these worsened health indicators translate into greater emergency room (ER) visits or broader healthcare utilization. This gap is important because obesity and diet related conditions such as diabetes and cardiovascular disease are known drivers of preventable and costly ER visits. From an economic perspective, this missing link matters for understanding consumption smoothing and utility maximization. For example, if individuals underweight long term health costs when making short term dietary decisions, the resulting increase in acute care usage may impose both private and public costs. Thus, while the literature convincingly establishes a causal pathway between fast food density and poorer health outcomes, it provides limited evidence on whether environmental exposure to fast food increases healthcare utilization, particularly ER visits. Addressing this gap would extend the causal chain from food environment to measurable strain on the healthcare system, strengthening both the policy and economic relevance of the research question. This study argues that higher county level fast food density increases preventable emergency department utilization among Medicaid eligible populations by worsening diet related health outcomes and barriers to preventive care.

Literature Review

The geographic distribution of fast food restaurants has become important in understanding diet related health differences in the United States. In many low income communities, fast food outlets and convenience stores tend to be more prevalent than grocery stores or other sources of healthy foods. These local conditions shape what foods are easiest to access daily and influence dietary behavior and long term health outcomes over time. Research on neighborhood food environments often shows that poorer communities generally face both limited access to healthy food stores and more exposure to cheaper yet calorically dense food options (Larson et al., 2009). As a result, the types of food outlets present in a community can impact consumption patterns and contribute to broader disparities in diet related diseases like obesity, diabetes, and cardiovascular conditions.

Areas where residents have limited access to supermarkets, fresh foods, or food deserts are primarily targeted in early studies on food access and among those most heavily impacted are low income regions. More recent work shows that the relative concentration of unhealthy options may matter more than the absence of healthy ones. Evidence demonstrates a lack of affordable, healthy food options (Larson et al., 2009) in the mentioned areas where environmental conditions (like transportation problems) only make access constraints worse. Counties considered food swamps, regions where unhealthy food outlets outnumber healthier alternatives, show stronger correlations with obesity rates than traditional food desert measures (Cooksey-Stowers et al., 2017). Researchers believe that the surrounding food environment impacts diet through exposure and convenience which allows us to analyze how local environments shape everyday food choices although using county level data makes it difficult to observe individual behavior.

Research examining the geographic distribution of food retailers further supports this perspective. Evidence reviewing national patterns of fast food locations shows that fast-food restaurants have the tendency to cluster in areas with higher population density and lower income levels (Fraser et al., 2010). This suggests a pattern that fast food establishments may disproportionately locate in neighborhoods where demand for inexpensive and convenient meals is high. As a result, residents of disadvantaged communities may face greater daily exposure to unhealthy food options than residents of higher income neighborhoods.

Other studies try to establish a causal relationship between the exposure of fast food and health outcomes. Currie et al. (2010) find that proximity to fast food stores increases obesity among students and greater weight gain among pregnant women. Through the comparison of individuals living extremely close to fast food restaurants and those living slightly farther away, the research is able to address the concern of fast food outlets establishing locations in areas with pre-existing unhealthy eating patterns. The results therefore suggest that proximity itself may influence health outcomes rather than slightly representing underlying preferences. Similar patterns are seen where shorter distance to fast food stores increases the probability of childhood obesity among residents of New York City public housing (Han et al., 2020). Differences in fast food proximity provide a useful source for identifying causal effects because housing placements are determined administratively in these developments.

Highlighting the role of convenience in shaping dietary decisions with evidence, research

on consumer behavior shows that even when individuals receive nutritional information about caloric content, they prioritize convenience (Wisdom et al., 2010). Convenience may therefore reinforce unhealthy dietary patterns in areas with a high concentration of fast food stores. Fast food stores in close proximity to homes, schools, or workplaces become easily accessible and a part of individuals' daily dietary routines. Over time, repeated exposure may normalize the consumption of highly processed foods and contribute to the emergence and development of diet related health conditions.

Financial limitations also have a significant influence on dietary decisions, apart from convenience. The cost of healthy foods may be a significant obstacle to better nutrition, especially for low income households, as indicated by research. Even when supermarkets are close by, healthier foods are still unaffordable and thus unattainable (Drewnowski et al., 2012). Because of this, people with low incomes might switch to less expensive, high calorie foods that are higher in calories per dollar but have less nutritional value. The correlation between food prices and consumption patterns explains why diet related health issues are more prevalent in lower income communities. The widespread availability of fast food options can make it a better alternative to healthier but more expensive options.

These financial limitations are especially important for households that rely on government food assistance programs. When household food resources become scarcer near the end of the monthly SNAP benefit cycle, emergency department visits rise, according to research on the timing of Supplemental Nutrition Assistance Program benefits (Cotti et al., 2020). This implies that short term health conditions may be impacted by changes in food access. Temporary food shortages or decreased nutritional intake can worsen pre-existing health conditions and necessitate emergency medical care. These results demonstrate how vulnerable populations may use healthcare more frequently because of food insecurity and financial instability. Related studies establish a direct connection between emergency department usage patterns and food insecurity. Individuals experiencing unstable access to adequate nutrition are significantly more likely to rely on emergency departments for medical care compared to individuals with consistent food access (Basu et al., 2017). Food insecurity can worsen chronic health conditions increasing the likelihood of acute health complications that require immediate treatment. Healthcare policy research also provides insight into how socioeconomic factors

influence patterns of emergency department use. Taubman et al. (2014), using a randomized health insurance experiment, show that individuals who gain Medicaid coverage increase their use of emergency department services. This increase partly reflects newly insured individuals receiving treatment for health conditions that were previously untreated due to lack of access to healthcare. But the findings also highlight the broader role that emergency departments play as a source of care for populations with limited access to primary care services. When preventive care is unavailable or difficult to access, individuals may rely on emergency departments to address health problems that could otherwise be treated in outpatient settings.

The geographic density of fast food restaurants and the use of emergency rooms for diet related ambulatory care sensitive conditions are not well correlated, despite these observations. Most of the literature currently in publication either looks at the factors that influence low income populations' use of emergency rooms or the connection between food environments and obesity. Nevertheless, few studies combine these two lines of inquiry to investigate whether unhealthy eating environments influence avoidable healthcare use. Understanding this relationship is very important for Medicaid populations who often experience both increased exposure to unhealthy eating environments and restricted access to preventive healthcare.

Conceptual Framework

This analysis is guided by a framework that connects local food environments to preventable health care utility. At the first stage, the locational concentration of fast food restaurants represent the relative cost of consuming unhealthy, nutrient-poor meals. Consumers typically substitute unhealthy, cheap options in the case of a low household income (Drewnowski et al., 2012; Wisdom et al., 2010). At the second stage, consistent habits and exposure can lead to the development of diet-related chronic conditions including obesity, type 2 diabetes, and cardiovascular disease (Currie et al., 2010; Han et al., 2020). At the third stage, these chronic conditions are not managed through outpatient care, leading to consumers having to increase utility of preventable emergency healthcare, particularly among populations with limited access to primary or preventative services.

Poverty is hypothesized to moderate this chain at multiple points. Low-income households face tighter budget constraints that increase substitution towards cheaper options that are typically higher in calories. They also experience greater barriers to preventative care that would interrupt the progression of chronic disease and rely more heavily on emergency departments for care (Cotti et al., 2020; Taubman et al., 2014). Together, these mechanisms imply that the marginal effect of fast food density on preventable ED utilization should be larger in high poverty counties than low ones.

This framework directly leads to our central thesis: higher county level fast food density increases preventable emergency department utilization among Medicaid-eligible populations by worsening diet-related health outcomes and compounding barriers to preventative care, with this effect amplified in high poverty counties. The empirical specification that follows is designed to test the implied predictions: a positive coefficient on fast food density, a positive coefficient on the high poverty indicator, a positive coefficient on their interaction, and a negative coefficient on median household income.

Empirical Methodology

Data

This study uses a county level dataset to examine the relationship between fast food density and preventable emergency department (ED) utilization for diet related ambulatory care sensitive conditions. For comparison purposes, cross sectional data is used, but our data structure is in the form of panel data. The reason for this is so we can extract more observations. There are 372 county observations covering the time period from 2011 through 2016. Healthcare utilization data is drawn from New York SPARCS, food environment measures are taken from the USDA Food Environment Atlas, and socioeconomic variables come from the U.S Census American Community Survey (5-year estimated) and the U.S Department of Housing and Urban Development Zip Code County Crosswalk figures. Together, this data is analyzed to assess how geographic access to fast food may influence health outcomes and healthcare use.

The primary variables of interest include a measure of preventable ED utilization (ppv_rate), which shows the rate of ED visits for conditions that are usually manageable through outpatient care. This variable is the dependent variable in the analysis and reflects differences in healthcare utilization across counties. The key explanatory variable is fast food density

(ffr_per1000) which is measured as the number of fast food restaurants per 1,000 people in a community. The dataset also includes a measure of full service restaurant density (fsr_per1000) and a ratio of fast food to full service restaurants (ffr_fsr_ratio) which provides an alternative way to show the balance between unhealthy and healthier food options in the community. There are also socioeconomic conditions such as the poverty rate (pct_poverty) which measures the percentage of individuals living below the federal poverty line in each county.

In interpreting the data, variation in fast food density and poverty rates is sufficient to estimate the relationship between food environments and ED utilization. Also, the inclusion of both continuous and categorical measures of socioeconomic status allows for a more nuanced analysis of how economic conditions interact with food access. The interaction between fast food density and high poverty status which can also introduce the possibility that the effect of fast food exposure may be stronger in more economically disadvantaged counties.

TABLE 1. Variable Descriptions

Variable Name	Description	Units/Measurement
ppv_rate	Rate of preventable ED visits for diet related ambulatory care sensitive conditions	Ratio (per population, 1000 residents)
ffr_per1000	Number of fast food restaurants per 1,000 residents	Restaurants per 1,000 residents
fsr_per1000	Number of full service restaurants per 1,000 residents	Restaurants per 1,000 residents
ffr_fsr_ratio	Ratio of fast food restaurants to full service restaurants	Ratio (fast food / full service)
pct_poverty	Percentage of county population living below federal poverty line	Percent (%)
high_poverty	Indicator for high poverty county	Dummy variable (1 = high poverty, 0 = otherwise)

median_hh_income	Median household income in the county	US dollars ($)
log_median_inc	Natural logarithm of median household income	Log transformed dollars
ffr_x_highpov	Interaction between fast food density and high poverty	Continuous (ffr_per1000 x high_poverty)

TABLE 2. Summary Statistics

	N	Mean	Std. Dev.	Min	Median	Max
ppv_rate	372	27.838	10.561	7.845	25.036	66.739
ffr_per1000	372	0.713	0.198	0.363	0.695	1.602
fsr_per1000	372	1.039	0.518	0.429	0.920	3.949
ffr_fsr_ratio	372	0.746	0.223	0.278	0.714	1.556
pct_poverty	372	13.089	3.436	5.300	13.050	26.900
high_poverty	372	0.500	0.501	0.000	0.500	1.000
median_hh_income	372	76432.694	18312.034	49036.000	70695.000	143408.000
log_median_inc	372	11.220	0.209	10.800	11.166	11.873
ffr_x_highpov	372	0.347	0.378	0.000	0.212	1.602

The values depicted in Table 2 show that there is meaningful variation across our dependent and explanatory variables, which supports the use of regression analysis in our study. For example, our dummy variable high_poverty has a mean of 0.500. This value represents a threshold defined at the sample median and illustrates that the sample is evenly split between

high and low poverty counties. Such variability will assist in our estimations regarding the relationship between fast food density and preventable ED utilization. In addition, high-poverty counties in our dataset exhibit a higher mean preventable ED visit rate (30.4 vs 25.3), which provides support for the expected positive relationship between healthcare utilization and poverty, while the negative correlation between pct_poverty and log_median_inc (r = -0.66), may introduce multicollinearity. Fast food is also positively correlated with full-service restaurants (r = 0.57), which helps us avoid OVB. One important data limitation is that USDA restaurant data is only available for 2016. Because food environments change at slow rates, data is limited. It is useful for 2016 to be a proxy year to combat this limitation. This renders our food environment measures time-invariant even though our data is structured as a panel. This means we are estimating a cross-sectional association rather than exploiting within-county variation over time. Because of this, we will employ a Pooled OLS estimator. This approach allows us to use the full 372 observations in our panel and improve the precision of our estimates, even though our primary identification of our parameters comes from cross-sectional differences. Other potential limitations include the unanticipated spread of ppv_rate values and the relatively limited variation of ffr_per1000 compared to other continuous variables.

Model Specification and Estimation Strategy

$$PPV_i = \beta_0 + \beta_1 FFR_i + \beta_2 HighPov_i + \beta_3 (FFR \times HighPov)_i + \beta_4 \ln lnInc_i + \beta_5 FSR_i + u_i$$

where PPV is the preventable visit rate, FFR is fast-food restaurants per 1,000 population, HighPov is the high-poverty indicator, lnInc is log median household income, and FSR is full-service restaurants per 1,000 population.

The dependent variable ppv_rate is the preventable emergency department (ED) visit rate for diet related ambulatory care sensitive conditions, at the county level. The main independent variable is ffr_per1000 which represents the number of fast food places per 1,000 people. The model includes a dummy variable, high_poverty, which is assigned to 1 if the county is high poverty and 0 if not- with lower poverty counties being in the omitted category. The interaction term ffr_x_highpov tests if the effect of fast food restaurants on preventable ED visits is magnified by high poverty. log_median_inc is controlled for in the model as the natural log of median household income, as is fsr_per1000, the number of full service restaurants per 1,000 people. For the coefficients to be interpreted as the marginal effect on the preventable ED visit

rate (caused by a change in an independent variable while holding other independent variables constant) , the model uses a linear functional form. Finally, to reduce skewness, the log of median household income is used, giving a more interpretable relationship between income and healthcare utilization.

On ffr_per1000, the expected sign is positive because fast food density is expected to worsen diet related health outcomes and also increase preventable emergency department use. Similarly, the expected sign on high_poverty is also positive since high poverty counties can face more barriers to preventive care and access to healthy foods. The coefficient on the interaction term, ffr_x_highpov, is also positive as the effect of fast food density may be stronger in higher poverty counties. Since higher income is usually associated with more or better access to preventive care and healthier food options, the expected sign on log_median_inc is negative. Lastly, the expected sign on fsr_per1000 is ambiguous but may be negative if having more full service restaurants means that people rely less on fast food.

Results

Table 3 presents Pooled OLS estimates of the relationship between fast-food density and preventable emergency department (ED) visit rates across 62 New York counties from 2011 to 2016. We present three specifications: Column 1 includes only our variable of interest, fast-food restaurant density per 1,000 population; Column 2 adds controls for full-service restaurant density and log median household income; and Column 3 is the full specification from our model. It adds the high-poverty county dummy and its interaction with fast-food density. Our key variables, fast-food and full-service restaurant density, are measured only in 2016 and are not varied across our panel years. This means that a fixed effects estimator would completely absorb these variables. We therefore employ a Pooled OLS. It identifies the relationship through cross-sectional variation across counties while still benefiting from the larger sample size of the panel.

Table 3. Pooled OLS Estimates: Fast-Food Density and Preventable ED Visit Rates

	(1)	(2)	(3)
Fast-food restaurants per 1,000 pop	−16.058***	−9.542***	−10.025**
	(2.642)	(3.513)	(4.869)
Full-service restaurants per 1,000 pop	—	2.107*	2.397**
		(1.177)	(1.205)
Log median household income	—	−18.464***	−15.888***
		(2.758)	(3.138)
High-poverty county (=1)	—	—	2.986
			(3.760)
FFR per 1,000 × High poverty	—	—	−1.434
			(5.070)
Constant	39.294***	239.630***	209.770***
	(1.956)	(29.964)	(34.370)
Observations	372	372	372
R^2	0.091	0.210	0.217
Adjusted R^2	0.088	0.203	0.206

Notes: Standard errors in parentheses. Dependent variable is the risk-adjusted preventable visit rate (PPV) per 100 people.
* p < 0.10, ** p < 0.05, *** p < 0.01
All models estimated using Pooled OLS on 62 New York counties, 2011–2016. Restaurant density measures are from the USDA Food Environment Atlas (2016). Income and poverty measures are from the American Community Survey (5-year estimates).

The coefficient on fast food restaurant density is negative and statistically significant across all specifications which is opposite of the hypothesized positive relationship. In the fully specified model, a one standard deviation increase in fast food restaurants per 1000 population (approx. 0.198 restaurants) is associated with a 1.987 point decrease in the risk adjusted PPV rate, representing about 7.1% of the sample mean of 27.84. That said, this result does not hold up under standard error assumptions. When standard errors are clustered at the county level the coefficient becomes statistically insignificant ($p = 0.241$). This suggests the negative association is not robust and the finding should be interpreted with caution.

The interaction term between fast food density and high poverty status is small and statistically insignificant ($\beta = -1.434$, $p = 0.778$), providing no evidence that the effect of fast food density on preventable emergency department visits differs between high and low poverty counties. This does not support the hypothesis that fast food access has a stronger adverse effect in poorer communities. A linear combination test finds that the total effect of fast food density in

high poverty counties is -11.458 (−10.025 + (−1.434)) which is statistically significant at the 1% level under conventional standard errors, $p = 0.005$. However, this result loses significance once standard errors are clustered at the county level, so it should be interpreted cautiously. Taken together, the interaction results suggest that poverty does not meaningfully condition the relationship between fast food density and preventable emergency department utilization at the county level.

While the coefficient of the high poverty dummy is positive yet insignificant ($\beta = 2.986$, p-value = 0.428), it is evident that high poverty areas have higher rates of preventable ED visits, but this effect diminishes when additional variables are considered in the regression analysis. Looking at the raw data, high-poverty: 30.4 vs. low-poverty: 25.3 indicates that high poverty counties have a higher average PPV rate. But this gap does not remain statistically meaningful after controlling for other factors like income and restaurant density.

Log median household income is the strongest and most robust predictor in our model and is statistically significant at the 1% level, $\beta = -15.888$, $p < 0.001$. This relationship remains statistically significant at the 1% level after clustering standard errors, $p = 0.004$, which helps to reinforce the consistency of the relationship across specifications. Since income is measured in logs, the coefficient here implies that a 1% increase in median household income is associated with a 0.16 point decrease in PPV rates. The coefficient on full service restaurants per 1,000 population is positive and statistically significant at the 5% level, $\beta = 2.397$, $p = 0.047$. This finding is somewhat unexpected given the expected relationship between restaurant availability and health outcomes. This result loses statistical significance once standard errors are clustered at the county level ($p = 0.468$), however, which indicates that the relationship is not robust.

The full model explains some of the differences in preventable emergency department (ED) use across counties. The R^2 is 0.217 and the adjusted R^2 is 0.206 meaning the model explains about 22% of the variation in PPV rates. The F statistic is 20.24 with $p < 0.001$ showing the model is statistically significant at the 1% level overall. We should, however, be careful when interpreting some of the results as shown by the diagnostic tests. The Breusch-Pagan test is also significant at the 1% level ($\chi^2 = 70.80$, $p < 0.001$) meaning there is heteroskedasticity and so supports using robust/clustered standard errors. There is high multicollinearity for high poverty

variables and interaction terms, per the VIF results (VIF = 14.85 and 15.42, respectively). This means the estimates should be interpreted carefully as they may be less precise.

The regression results do not support our original hypothesis that stated more fast food restaurants were linked to higher preventable ED use. We learned that fast food density had a negative coefficient in each model but the result is not strong once clustered standard errors are used. There is no clear evidence that the relationship is stronger in high poverty counties as the interaction between fast food density and high poverty status is not statistically significant. In contrast, the median household income was the most consistent predictor in the model showing lower PPV rates with higher income counties. The results suggest that county level income matters more for preventable ED use than fast food restaurant density.

Discussion and Conclusion

According to our findings, fast food restaurant density does not appear to meaningfully predict preventable emergency department utilization once we take socioeconomic factors into account. While the coefficient on fast food restaurant density is negative (FFR = −10.025), the relationship loses significance once standard errors are clustered at the county level. As a result, we are unable to identify a causal effect. In addition, the interaction between fast-food density and poverty status is statistically insignificant, suggesting that the relationship between fast-food density and ED utilization does not differ meaningfully between high- and low-poverty counties. Conversely, log median household income is the only variable in our study that remains both statistically significant and robust across all model specifications. The negative coefficient indicates that higher income counties experience lower rates of preventable ED utilization, which suggests that small increases in income are associated with meaningful reductions in PPV rates. This finding is logical, as higher income populations likely have improved access to healthcare resources, including both primary and preventative care, reducing reliance on emergency department use. Our results thus suggest that, at the county level, broader economic conditions play a significant role in shaping preventable ED, and that the food environment does not predict preventable ED once income is controlled.

While the negative coefficient on fast food restaurant density, given prior research (Currie et al. 2010, Han et al. 2020), is unexpected, several logical explanations can be provided to bring

meaning to our results. Firstly, fast food density may serve as an indicator of overall economic development rather than directly reflecting dietary habits. Counties with higher density ratings tend to be those that are the most urbanized. As a result, they are also likely the most wealthy, which is reflected in the positive correlation between fast food density and median household income. With such wealth also comes greater access to healthcare infrastructure, which can reduce emergency department usage. Therefore, the negative association may be capturing differences in healthcare access and utilization patterns rather than stemming from the direct influence of fast-food consumption itself.

Secondly, rural communities, or those with fewer restaurants of all types are less populated when compared to urban areas, and additionally have limited healthcare access. This limited access to care is associated with higher rates of preventable ED utilization, and thus the negative coefficient may be capturing rural-urban divide, rather than a causal relationship between health outcomes and fast food availability.

Finally, our analysis faces an ecological fallacy where the use of county-level data prevents us from identifying whether individuals who consume more fast food are the same individuals driving ED utilization. In previous studies regarding fast food density and health outcomes, more granular data have found positive effects of fast food exposure on health outcomes (Currie et al. 2010, Han et al. 2020). This suggests that our usage of county-level aggregated data may be obscuring the true relationship. Additionally, there may be potential bias in the estimated relationship as fast food restaurant placement is not randomized. Firms develop in areas where demand is typically strong and income is high, thus introducing endogeneity into the model rather reflecting the independent effect of fast food access itself.

The interaction between fast food density and high poverty is small and not statistically significant ($\beta = -1.434$, $p = 0.778$). There is no evidence that fast food density affects high poverty counties compared to low poverty ones. This is also because log median income already captures most of what poverty is doing in the model. Income has a strong negative relationship with preventable ED visits ($\beta = -15.888$, $p < 0.001$), so once it's included, the high poverty dummy does not add much new information. The high poverty variable and interaction term also move closely together, which presents a precision issue. This shows up in high VIF values, both around 15. This makes it hard to separate individual effects. Standard errors get larger and even

if there were a real difference, the model would struggle to detect it. So the lack of significance does not fully rule out a heterogeneous effect.

These results differ from Currie et al. (2010) and Han et al. (2020), who find that fast food exposure increases obesity and related health outcomes. Their studies use individual or highly localized data and measure proximity within very small distances, which captures actual behavior. In our model, fast food density is measured at the county level as restaurants per 1,000 residents. That aggregation smooths over within county differences so the mechanism they identify is harder to detect. This explains why our coefficient on fast food density is negative (-10.025) and not robust once we cluster standard errors, instead of positive as expected.

The results align more with Basu et al. (2017) and Cotti et al. (2020), which focus on economic constraints as a driver of emergency department use. In our model, log median household income has a large and statistically significant coefficient ($\beta = -15.888$, $p < 0.001$) and remains significant with clustered standard errors. Higher income counties have lower rates of preventable ED visits. Once income is included, fast food density does not meaningfully predict ED use, which points to economic conditions as the main factor at this level of analysis.

Although our research is framed in causal terms, our results should be interpreted as associations instead of causal effects. A reason is that fast food restaurants are not assigned random locations across counties. The density of restaurants may be influenced by factors such as income, population density, development, or local demand making the fast food variable endogenous. Also, a fixed effects model does not work with our main explanatory variables which are time invariant; it would absorb those county level variables entirely. As a result, we are not able to control for unobserved county characteristics the same way a stronger causal design would. For these reasons, our analysis does not identify the causal effect of fast food density on preventable ED use, but it estimates county level associations instead.

Our results suggest that policies focused on income can be more helpful for reducing preventable ED use than policies focused only on fast food access. Since median household income was the strongest and most consistent predictor in our model, we conclude that lowering poverty or improving access to healthcare (Medicaid) matters more than changing the food environment at the county level. Targeting fast food access in those counties may not reduce ED visits as we did not find evidence that fast food density had a stronger effect in high poverty

counties. This still does not mean food environments do not matter. Because our data is at the county level, future research with ZIP -code or individual level data is needed before we can make stronger conclusions.

There are several limitations that should be considered when interpreting the results. First, the food environment data is time-invariant. Restaurants are drawn from a single year, 2016, and applied across the full study period from 2011 to 2016. As a result, there is no within county variation in fast food density over time. Another limitation is the use of a coarse unit of analysis. The data is aggregated at the county level which may hide a lot of variation within counties. Large or diverse counties can have very different food environments, income levels, and access to healthcare depending on certain neighborhoods within that county. If ZIP code or individual level was used instead, it could have given a more specific relationship comparison. A third limitation is omitted variable bias due to several relevant factors not included. The analysis does not include controls for healthcare supply like primary care providers per capita or hospital bed availability which could influence emergency department use separately from food environments. Another limitation is endogeneity. Because locations of fast food restaurants are not random, other unobserved factors such as local demand, population density, or consumer preferences could be influencing the analysis. It is hard to know whether fast food density directly affects emergency department utilization or whether both are driven by underlying characteristics without the use of an instrumental variable or natural experiment. Finally, the scope of the data is limited geographically. The sample consists of 372 observations from 62 counties within only New York. There could be differences in food environments, healthcare systems, and population characteristics across the United States that this sample is not taken into consideration. As a result, the conclusions should be interpreted as context specific rather than universally applicable.

Our research question is: Does county level fast food density affect preventable emergency department utilization among Medicaid eligible populations? We find that the results do not support the hypothesized positive relationship between fast food density and preventable emergency department utilization use. We also find that the expected amplification effect in high poverty counties that we hypothesized is also not supported. Instead, fast food restaurant density is negatively associated with preventable visit rates and statistically insignificant when using

clustered standard errors. While the primary variable of interest is insignificant, the analysis does find that median household income is the most dominant predictor of preventable emergency department utilization. Higher income counties have substantially lower rates of preventable emergency department visits. This finding suggests that socioeconomic factors rather than the local concentration of fast food restaurants plays a role in influencing healthcare utilization. For future work, studies should use individual level or ZIP code level data which would allow for a more precise understanding of how exposure to fast food influences dietary choices and healthcare utilization. Despite the null FFR result, the income finding has clear implications. Efforts to reduce poverty may be a more effective way to lower preventable emergency department use among low income populations than interventions on the food environment.

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